

# REACT INTERVIEW QUESTIONS :

## 1. What is React.js ?

**Ans :** React.js is a JavaScript library for building user interfaces. It allows developers to create reusable UI components that update efficiently.

## 2. Explain the concept of Virtual DOM in React

**Ans :** The Virtual DOM is a lightweight copy of the actual DOM. React uses it to improve performance by minimizing direct manipulation of the real DOM, updating only the necessary parts.

## 3. What are React components?

**Ans :** React components are building blocks of a React application, representing UI elements. They can be class components or functional components.

## 4. What is JSX?

**Ans :** JSX (JavaScript XML) is a syntax extension for JavaScript recommended by React. It allows writing HTML elements and components in a syntax similar to XML or HTML.

## 5. Explain the difference between state and props.

**Ans :** Props (short for properties) are inputs to a React component, while state is an internal data storage mechanism. Props are immutable, and changes to them trigger a re-render, while state changes trigger a component update.

**6. What is the significance of setState() in React?**

**Ans :** setState() is used to update the state of a React component. It triggers a re-render and ensures that the updated state is reflected in the rendered UI.

**7. How does React handle user input?**

**Ans :** React handles user input through controlled components, where the component state is used to manage the input fields, and the UI is updated based on the state.

**8. Explain the concept of lifting state up in React.**

**Ans :** Lifting state up involves moving the state of a child component to its parent to share the state between components. This is often done to maintain a single source of truth

**9. What are React hooks?**

**Ans :** React hooks are functions that enable functional components to use state and lifecycle features. Examples include useState and useEffect.

**10. What is the purpose of the useEffect hook?**

**Ans :** useEffect is used for side effects in functional components, such as data fetching, subscriptions, or manually changing the DOM.

**11. What is Next.js?**

**Ans :** Next.js is a React framework for building server-rendered React applications with features like automatic code splitting, server-side rendering, and easy configuration.

**12. Explain the difference between client-side rendering (CSR) and server-side rendering (SSR) in Next.js.**

**Ans :** CSR renders pages on the client side after the initial page load, while SSR renders pages on the server side and sends HTML to the client.

**13. How does Next.js handle routing?**

**Ans :** Next.js uses a file-based routing system. Files in the ``pages`` directory automatically become routes, and nested folders represent nested routes.

**14. What is the purpose of `getInitialProps` in Next.js?**

**Ans :** ``getInitialProps`` is used to fetch data on the server side and pass it as props to a page component. It is commonly used in SSR.

**15. Explain the concept of dynamic routing in Next.js.**

**Ans :** Dynamic routing allows creating pages with dynamic content based on the URL. Parameters are defined in the file name using brackets (e.g., ``[id].js``).

**16. How does Next.js handle API routes?**

**Ans :** Next.js has a built-in API routes feature that allows creating API endpoints within the ``pages/api`` directory. These endpoints are serverless functions.

**17. What is serverless deployment in the context of Next.js?**

**Ans :** Serverless deployment involves deploying Next.js applications to serverless platforms like Vercel or Netlify, where the server setup and scaling are abstracted.

**18. Explain the purpose of the `next/head` component**

**Ans :** `next/head` is used to modify the head of a page. It allows adding metadata, links to stylesheets, and other elements to the HTML head.

**19. How does Next.js support internationalization (i18n)?**

**Ans :** Next.js has built-in support for i18n, allowing the creation of internationalized applications by providing translations and locale-specific routes.

**20. What are the benefits of using Next.js over a standard React application?**

**Ans :** Next.js simplifies server-side rendering, routing, and deployment, offering a more opinionated framework for React that enhances performance and developer experience.

**21. How does React differ from other JavaScript frameworks/libraries?**

**Ans :** React focuses on building UI components and employs a virtual DOM for efficient updates, allowing for a more modular and declarative approach to building user interfaces.

**22. Explain the concept of "lifting state up" in React, and when would you use it?**

**Ans :** Lifting state up involves moving the state of a child component to its parent to share the state between components. This is often done to maintain a single source of truth

**23. What are React hooks, and why were they introduced?**

**Ans :** React hooks are functions introduced in React 16.8 to allow functional components to use state and lifecycle features, making it easier to reuse stateful logic.

**24. How does React handle conditional rendering? Provide examples.**

**Ans :** React handles conditional rendering using JavaScript expressions. For example, using the ternary operator (`condition ? true : false`) or the logical AND (`&&`) operator.

**25. What are the key differences between class components and functional components in React?**

**Ans :** Class components use ES6 classes and have a state and lifecycle methods, while functional components are simpler, stateless, and use hooks for state and side effects.

**26. Explain the concept of React props drilling and how to avoid it.**

**Ans :** Props drilling occurs when props need to be passed through multiple layers of components. To avoid it, use React context or consider lifting state up.

**27. How does React handle lists and keys in rendering components?**

**Ans :** React uses the `map` function to render lists of components, and each component in the list should have a unique "key" prop to help React identify and update the components efficiently.

**28. What are higher-order components (HOCs) in React?**

**Ans : HOCs are functions that take a component and return a new component with additional props or behavior. They are used for code reuse and abstraction of logic.**

**29. Explain the concept of controlled and uncontrolled components in React forms.**

**Ans : Controlled components have their state managed by React, while uncontrolled components manage their state through the DOM. Controlled components are preferred for better control and predictability.**

**30. How does React handle state management in large-scale applications, and what external libraries can be used for this purpose?**

**Ans : React provides its state management using the `useState` and `useReducer` hooks. For larger applications, external**

**31. What is the virtual DOM?**

**Ans : DOM stands for Document Object Model. The DOM represents an HTML document with a logical tree structure. Each branch of the tree ends in a node, and each node contains objects. React keeps a lightweight representation of the real DOM in the memory, and that is known as the virtual DOM. When the state of an object changes, the virtual DOM changes only that object in the real DOM, rather than updating all the objects**

**32. Why is there a need for using keys in Lists?**

**Ans : A key is a unique identifier and it is used to identify which items have changed, been updated or deleted from the lists. It also helps to determine which components need to be rerendered instead of**

re-rendering all the components every time. Therefore, it increases performance, as only the updated components are re-rendered

**33. What is the use of render() in React?**

**Ans :** It is required for each component to have a render() function. This function returns the HTML, which is to be displayed in the component

If you need to render more than one element, all of the elements must be inside one parent tag like <div> <form>

**34. What is StrictMode in React?**

**Ans :** React's StrictMode is sort of a helper component that will help you write better react components, you can wrap a set of components with and it'll basically:

Verify that the components inside are following some of the recommended practices and warn you if not in the console.

Verify the deprecated methods are not being used, and if they're used strict mode will warn you in the console.

Help you prevent some side effects by identifying potential risks.

**35. Why do class methods need to be bound to a class instance?**

**Ans :** In JavaScript, the value of this changes depending on the current context. Within React class component methods, developers normally expect this to refer to the current instance of a component, so it is necessary to bind these methods to the instance. Normally this is done in the constructor—for example:

**36. WHEN WOULD YOU USE A CLASS COMPONENT OVER A FUNCTIONAL COMPONENT?**

**Ans : React uses two kinds of components—class components and functional components.**

**Functional components are the most basic kind of React component, defined by the components (unchanging) props.**

**Class components are more complicated React components that allow developers to execute component lifecycle methods and manage a component's state.**

**This means class components are used over functional components when you need to manage state or use component lifecycle methods**

### **37. WHAT ARE SOME OF THE MAJOR ADVANTAGES TO USING REACT WHEN BUILDING UIS?**

**Ans : Increased application performance via the Virtual DOM model**

**Improved coding efficiency with JSX**

**The ability to reuse components across multiple projects**

**Flexibility and extensibility through add-on tools provided by React's open source community**

### **38. WHAT ARE SOME OF REACT'S LIMITATIONS?**

**Ans : React JS is a JavaScript library and not a full-blown framework, meaning it might not be full service enough for some project**

**React's library is extremely large and can take additional time and experience (past the initial learning phase) to really understand**

**JSX adds a new coding dimension for developers who haven't used before (though it does have a gentle learning curve due to its similarity to HTML)**



**39. What are React Extensions? Name a few of them**

**Ans : React is more than just a UI framework; it contains a slew of extensions that cover the entire application architecture.**

**It has a server-side rendering feature.**

**Supports the creation of mobile apps.**

**Flux and Redux, among other things, have been added to the mix.**

**Flux is Facebook's application architecture for developing online applications.**

**React Native allows you to create mobile apps entirely in JavaScript**

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**React Assignment for practise :**

- 1. Create a simple React component that displays "Hello, World!" on the screen.**
- 2. Build a React component that displays a list of items fetched from an external API.**
- 3. Create a responsive navigation bar with React.**
- 4. Build a React form that allows users to enter their contact information.**
- 5. Implement client-side routing in a React application.**
- 6. Build a React component that displays a grid of images.**
- 7. Create a React component that displays a modal window when a button is clicked.**
- 8. Build a React component that implements a shopping cart.**
- 9. Implement state management in a React application using Redux.**
- 10. Create a React component that displays a countdown timer.**

- 11. Build a React component that implements pagination for a list of items.**
- 12. Create a React component that implements a slider or carousel.**
- 13. Build a React component that displays a map using Google Maps API.**
- 14. Implement server-side rendering in a React application.**
- 15. Create a React component that allows users to upload and display images.**
- 16. Build a React component that implements drag and drop functionality.**
- 17. Create a React component that displays a video player.**
- 18. Build a React component that implements infinite scrolling for a list of items.**
- 19. Implement authentication and authorization in a React application.**
- 20. Create a React component that allows users to filter and sort a list of items.**